

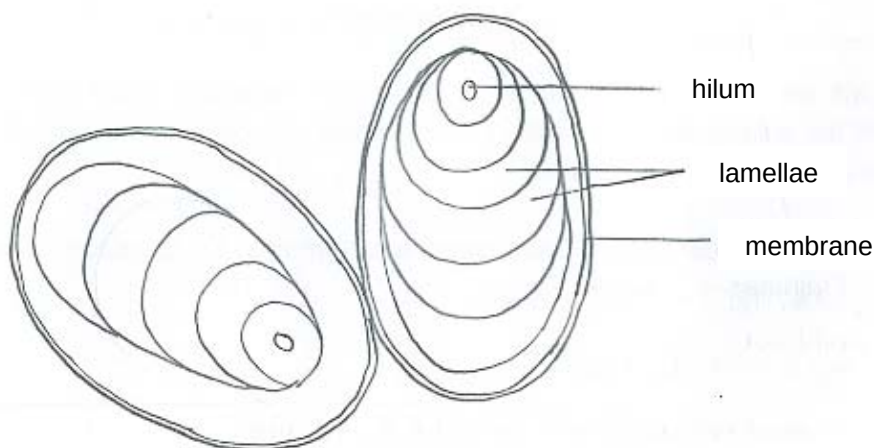


Question 1

3 Colour of extract on adding iodine solution [1]
blue-black

4 Make a large labelled detailed drawing of two adjacent starch grains. [4]

High power / detailed drawing of 2 adjacent starch grains from potato,
Solanum tuberosum (whole mount, 400x)



1. Drawing quality that includes clear continuous lines with no shading
2. Draw 2 adjacent starch grains
3. Correct shape of starch grains
4. Correct drawing of lamellae and hilum
5. All 3 correct labels: membrane, lamellae, hilum

5 Explain the result obtained in step 3 when iodine solution was added to the re-suspended T. [1]

1. crushing ruptures potato cells / cell walls / releases starch grains

OR

2. starch grains / amyloplasts are too large to pass through sieve and remain in residue / re-suspended T thus turning iodine solution blue-black.

12 Calculate the time taken to reach **each** end-point (**processed** results).

(a) Prepare the space below to record your results for:

- the test for starch
- the **raw** results for the appearance of glucose
- your **processed** results for the appearance of glucose.

[5]

time / min A: s	test for starch	test for glucose	
		time shown on stopwatch when end-point is reached	time taken to reach end- point / s
3	blue-black	4 min 17 s	77
6	blue-black	6 min 54 s	54
9	blue-black	9 min 50 s	50
12	blue-black	12 min 32 s	32
15	yellowish-brown	15 min 30 s	30

1. correct choice of equal time intervals which must span entire 15 min
2. correct column headings and units: time / min, test for starch, time shown on stopwatch, time taken to reach end-point / s
3. appropriate colours recorded for starch test for at least four times;
4. correct pattern of results for glucose test – time taken to reach end-point decreases with increasing time
5. processed times recorded as whole seconds

(b) Account for the results obtained in **(a)**.

[3]

1. as time increases, a greater proportion of / more starch is broken down by starch phosphorylase into glucose phosphate
2. resulting in an increase in glucose phosphate concentration which leads to a shorter time taken to decolourise the potassium manganate (VII) solution / reach the colourless end-point
3. until all starch is broken down into glucose phosphate at the end of 15 minutes which leads to the iodine solution remaining yellowish-brown / negative results for starch test, indicating absence of starch

- (c) Identify **one** limitation of this investigation and suggest a way in which the experiment can be improved to give more accurate and reliable results. [2]

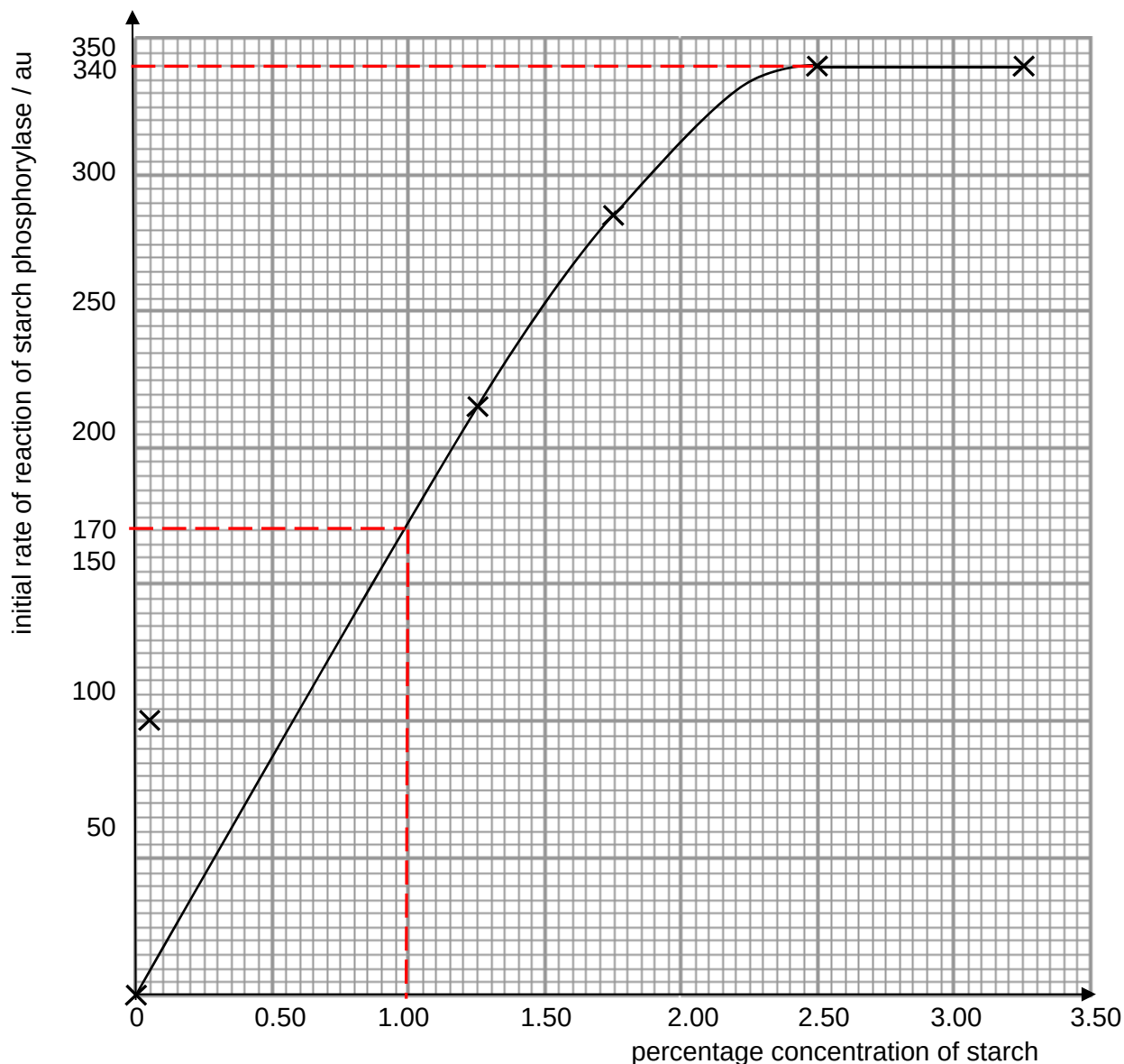
Limitation	Method to overcome limitation
L1. Use of naked eye to observe colour of achromic point / end-point OR visual comparison is subjective, resulting in inaccurate determination of colour of achromic point / end-point	M1. Use a colourimeter / spectrophotometer to determine colour intensity of reaction mixture
L2. Use of dropper / plastic pipette to withdraw reaction mixture OR different size / volume of drops of iodine solution / reaction mixture could contribute to differences in colour intensity for starch test	M2. Use a micropipette to withdraw a fixed precise volume of iodine solution / reaction mixture
L3. The contents of tubes were mixed manually OR rigour of manual shaking may be inconsistent, leading to inconsistent mixing of the tubes contents	M3. Use a mechanical shaker / vortex to mix the contents of tubes thoroughly to ensure a homogenous mixture;

- (d) Consider how you could modify this procedure to investigate the effect of **pH** on the activity of amylase. Describe how the independent variable, **pH**, could be investigated. [3]
1. at least five pH with stated values
 2. use of buffers;
 3. remove sample after set time / example of time and test with iodine and idea of looking for a colour change / test with potassium manganate (VII) and time taken to decolourise

(e)(i) Use the grid provided to plot a graph of the data shown in Table 1.1.

[4]

Effect of percentage starch concentration on initial rate of reaction of starch phosphorylase / au



1. correct choice of axes with independent variable (percentage concentration of starch) on x-axis
2. both axes correctly labelled including units
3. axes scaled appropriately so that graph takes up at least 50% of the grid and divisions are equidistant
4. Correctly plotted points and points joined by appropriate line of best fit (curve) as required by the data, showing plateau from 2.50% starch onwards, with no extrapolation beyond extreme measured data

(ii) Use the graph to estimate the Michaelis-Menten constant (K_m).

Show your working on the graph and in the space below.

[2]

1. shows on the graph V_{max} line at top of curve to the y-axis from the maximum rate of reaction
2. shows on the graph how K_m is read off at half V_{max}
3. correct answer for K_m from graph with correct units

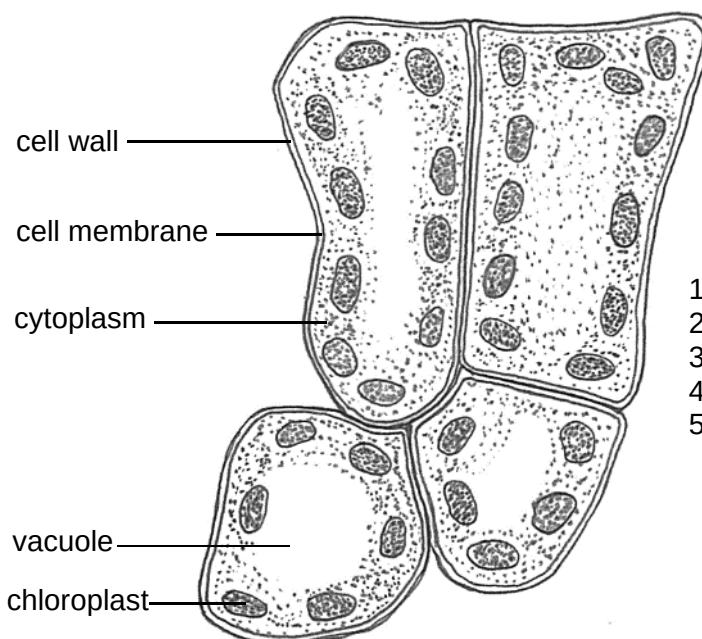
[Total: 25]

Question 2

(a)(i) Make a detailed, labelled drawing in the space below of this group of four cells.

[5]

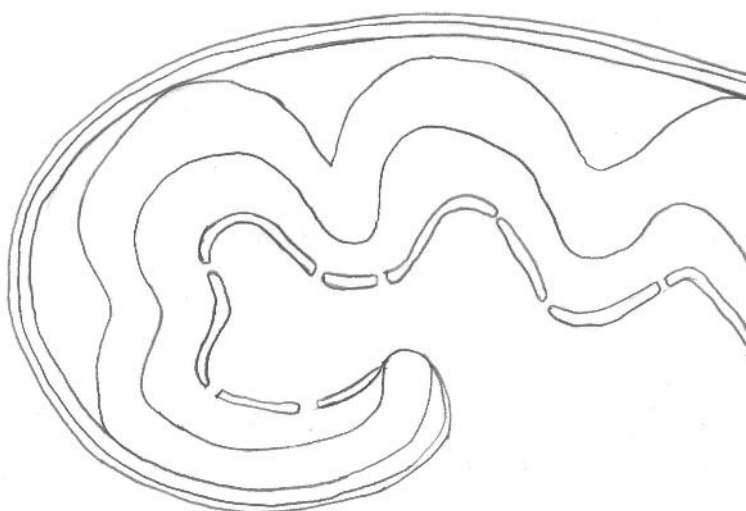
Detailed drawing of four cells found immediately below the upper epidermis (t.s., $\times 400$)



1. Accurate title
2. Quality of drawing
3. Position of cells
4. Shape and proportion of cells
5. Correct labels

(ii) Examine slide **S1** under the low-power objective lens of your microscope again. Make a large, plan drawing of the part of the leaf indicated by the shaded area in Fig. 2.1. On your diagram, use **one** ruled label line and label to identify one feature that adapts the plant to living in a dry habitat. Annotate this label to explain **how** the feature you have identified adapts this plant to living in a dry habitat.

[3]



1. Correct number and proportion of layers
2. Curving of lamina
3. Correct label and annotation

- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual width, in μm , of the leaf.

Show the measurements that you made and your working. [2]

1. Divide stage micrometer measurement by number of eyepiece graticule divisions
2. Measure width of leaf in eyepiece graticule divisions
3. Correct working and final answer in μm

- (b) Show the measurement that you made and your working for the width at position Q. [2]

P 577 μm , Q 697 μm , R 726 μm

1. Measure width of leaf
2. Divide by 175 and multiply by 1000
3. Correct answer to appropriate degree of accuracy

- (c) (i) State how narrower leaves allow xerophytic plants to conserve water. [1]

Narrower leaves presents a lower surface area

- (ii) Comment on what these results show and explain if the results support the student's hypothesis. [3]

1. Leaves of xerophytic plant species Y ($562.0 \mu\text{m}$) are significantly narrower than those of mesophytic plant species X ($1012.5 \mu\text{m}$)
2. Range of width of leaves of plant species X is not overlapping with that of plant species Y
3. Ref. to larger standard deviation as a larger spread of values around the mean width for leaves of plant species X compared with those of plant species Y.

[Total: 16]

Question 3

1. Ref to different organisms have different rates of respiration
2. ATP is synthesised with the production of carbon dioxide and water
3. Higher rate of respiration, the more oxygen consumed per unit time
4. Measure about 5 g of each organisms
5. Using a respirometer, measure the distance moved by the meniscus in a fixed time
6. Use petroleum jelly to seal between the graduated tube and the airtight container
7. Incubate respirometer in the water trough for 5 min
8. Setup equilibrate at room temperature for 1 minute
9. Repeat experiment for the other two organisms
10. Measure distance moved by meniscus
11. Calculate rate of respiration
12. Carbon dioxide absorbent corrosive / irritant, wear gloves and goggles to protect oneself.
13. AVP

[Total: 16]